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|------------------|----|
| No. of Pages     | 3  |
| No. of Questions | 3  |
| Total Marks      | 30 |
| Time: 90 minutes |    |

**Department of Computer Science and Engineering**  
**FINAL EXAMINATION Spring 2023**  
**CSE 110: Programming Language I**

- ✓ Write theory teacher's **Initial** and **SET number** on top of the answer script.
- ✓ Answer all questions. Use **back part** of the answer script for rough work.
- ✓ Answer Question **1 & 2** at the **beginning part** of the answer script.
- ✓ Figure in bracket [] next to each question indicates marks for that question.
- ✓ At the end of exam, put **question paper** inside answer script and **return both**.
- ✓ Understanding the question is part of the exam, **please do not ask questions**.

No washroom breaks

Section: \_\_\_\_ ID: \_\_\_\_\_ Name in CAPITAL: \_\_\_\_\_

**Question 1 [CO1] [10 Points]**  
**[Answer on the answer-script]**

Write a python program to generate and print a dictionary from a given list of marks for different tasks, where each key in the dictionary represents a task number ("task1", "task2" etc.) and the corresponding value represents the mark obtained for that task. Additionally, the program should calculate the average mark and include it in the dictionary as a key-value pair.

**NB:** You do not need to take list input from the user but your program should work for any given list of marks.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Given List 1:</b><br>marks = [85, 92, 78, 88]                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Given List 2:</b><br>marks = [80, 55, 92, 80, 79, 88]                                                                     |
| <b>Sample Output 1:</b><br>{ "task1": 85, "task2": 92, "task3": 78, "task4": 88, "average": 85.75 }                                                                                                                                                                                                                                                                                                                                                                 | <b>Sample Output 2:</b><br>{ "task1": 80, "task2": 55, "task3": 92, "task4": 80, "task5": 79, "task6": 88, "average": 79.0 } |
| <b>Explanation:</b><br>The key "task1" represents the first task which is the first element of the given 'marks' list. Similarly, the key "task2" represents the second task, which is the second element of the 'marks' list, and so on. Additionally, the key "average" represents the average mark which in the case of 'Given List 1', the average mark is $(85 + 92 + 78 + 88) / 4 = 85.75$ and $(80 + 55 + 92 + 80 + 79 + 88) / 6 = 79.0$ for 'Given list 2'. |                                                                                                                              |

## Question 2 [CO1] [10 Points]

[Answer on the answer-script]

Write a python function which takes a parameter string as name and **returns True** if the name is valid, otherwise **returns False**.

A name can consist of multiple words. The first character of each word should be in uppercase and the rest of the characters should be in lowercase. You may use `split()` function.

[Please check the next page for function call and sample outputs]

### Function Call 1:

```
print(checkname("Mr Alex Ben"))
```

### Sample Output 1:

True

### Explanation 1:

Here, each word of the name starts with an uppercase character and rest of the characters are in lowercase. Therefore, the function returns True.

### Function Call 2:

```
print(checkname("Mr AlEx ben"))
```

### Sample Output 2:

False

### Explanation 2:

Here, the 2nd word contains an uppercase character in the middle and the 3rd word of the name starts with a lowercase character which doesn't follow the perfect naming convention. Therefore, the function returns False.

### Question 3 [CO4] [10 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

|    |                                                |        |
|----|------------------------------------------------|--------|
| 1  | <code>list1 = [0, 2, 4, 6, 8]</code>           | Output |
| 2  | <code>list2 = ["H", "A", "P", "P", "Y"]</code> |        |
| 3  | <code>p = 34</code>                            |        |
| 4  | <code>q = 0</code>                             |        |
| 5  | <code>test = ""</code>                         |        |
| 6  | <code>while p &lt; 39:</code>                  |        |
| 7  | <code>test = test + str(p - q)</code>          |        |
| 8  | <code>q = (p - q + 9) % 3</code>               |        |
| 9  | <code>list1[q] = int(test) - list1[q]</code>   |        |
| 10 | <code>print(list1[q])</code>                   |        |
| 11 | <code>if q != 0:</code>                        |        |
| 12 | <code>test = str(p % q + 2)</code>             |        |
| 13 | <code>else:</code>                             |        |
| 14 | <code>test = str(3 - p * q)</code>             |        |
| 15 | <code>list2[q] = list2[q] + test</code>        |        |
| 16 | <code>print(list2[q])</code>                   |        |
| 17 | <code>p += 1</code>                            |        |
| 18 | <code>test = ""</code>                         |        |